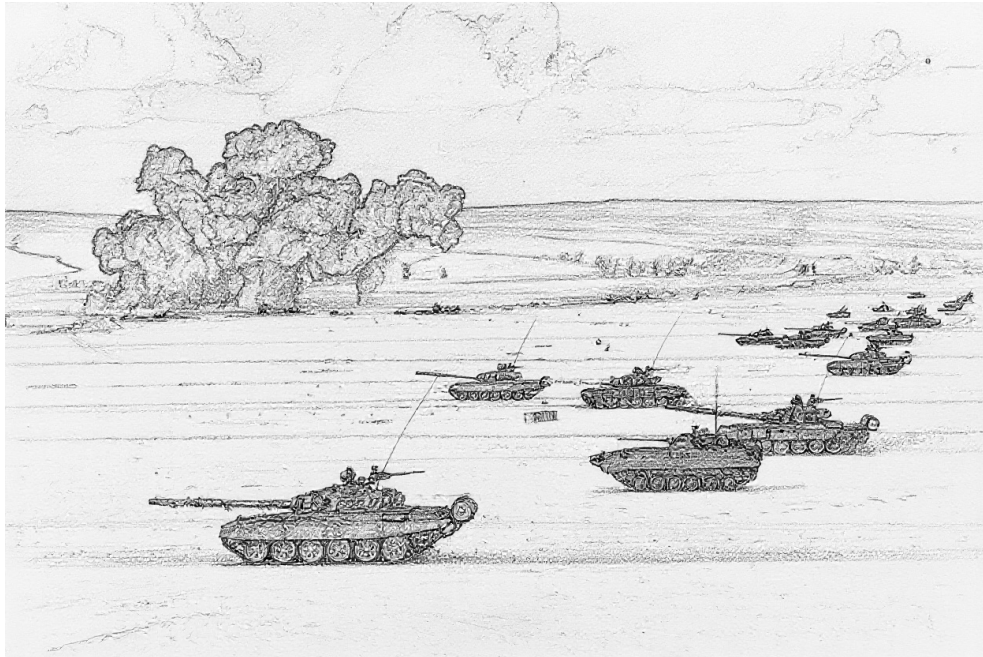


Air Power: Ground Unit and Ground Target Data

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1 May 2026



INTRODUCTION

For many, the principal focus of the *Air Power* game is air-to-air combat. That said, no less an authority on air-to-air combat than Robin Olds stated:¹

“You’re not going to win a war by shooting down a damn MiG, you’re going to win a war by bombing the bejesus out of whatever you were sent to bomb.”

Here we deal with those “whatevers,” the ground units and ground targets that are the targets of air-to-ground attacks, and some of which shoot back.

This document first presents ground-combat and transport units, then air-defense units (AAA, FCR, SAM, EWR, and CCU units), and finally ground targets. An appendix gives some examples of battalion-sized units.

Much of the data here is adapted from *Air Strike*, *Eagles of the Gulf*, and *The Speed of Heat*, from *Air Power Journal*, and from Malcolm Pipe’s extensive compendium of AAA and SAM units. However, the data have been carefully revised to improve their accuracy and internal consistency, a wider variety of ground-combat units are covered to give a richer experience, and the descriptions of the units and their weapon systems have been expanded.

Extensive notes accompany the main text, providing provenance of data, explanations of changes, sources of information, and justifications for decisions.

CONTENTS

GROUND-COMBAT AND TRANSPORT UNITS

Ground-combat units are formations of infantry, tanks, IFVs, APCs, other AFVs, or artillery whose primary role is combat with other ground units. Transport units provide strategic mobility to infantry and towed artillery and are the life-blood of logistic networks. These units are presented below with their counters. The Ground-Combat and Transport Unit Table summarizes their properties. Some representative examples of vehicles are given in the descriptions, and Vehicle Types Table gives a more extensive list. All ground-combat and transport platoons and batteries are also available as sections and squads.²

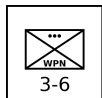
INFANTRY UNITS

Infantry are soldiers trained to fight on foot and, in some cases, from IFVs. Their primary roles in modern warfare are seizing, clearing, and holding ground, often as part of a combined-arms formation with tanks and artillery, highly mobile offensive and defensive operations in combination with helicopters, and combat in difficult terrain, including urban areas, mountainous zones, and dense jungle or forest.

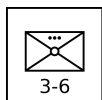
An important secondary characteristic of infantry is any associated transport. Motorized infantry units are transported by trucks, giving them operational mobility. Mechanized infantry are transported by APCs or IFVs, giving them both tactical and operational mobility. Light infantry units have no or fewer supporting vehicles, although modern light infantry is often airborne.



Infantry Platoon: An infantry platoon is armed principally with small arms, often supplemented with light machine guns and sometimes with light grenade launchers and portable anti-tank weapons such as bazookas, LAWs, MAWs, and portable ATGMs. It is the primary combat unit in larger infantry formations.

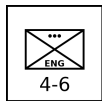


Infantry Weapons Platoon: An infantry weapons platoon is equipped with heavier weapons, such as medium or heavy machine guns, recoilless rifles, portable ATGMs, and automatic grenade launchers. It is often a company- or battalion-level asset, and its role is to provide direct-fire support to the infantry platoons in its formation.³

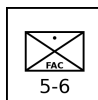


Infantry Mortar Platoon: An infantry mortar platoon is equipped with 60-mm, 81-mm, 82-mm, or similar mortars. It is often battalion-level assets, and its role is to provide indirect fire support and smokescreens to the infantry platoons in its formation.

Due to the weight of ammunition required for sustained fire, it will often be transported by a light truck platoon even if the other infantry units in its formation do not have transport.

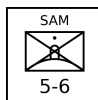


Infantry Engineers Platoon: An infantry engineers platoon is equipped and trained for combat engineering tasks such as mine-laying and mine-clearing. It is often battalion-level asset.



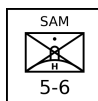
Infantry FAQ Squad: An infantry FAC squad has a forward air controller with other soldiers providing communication and protection.

An infantry FAC squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.



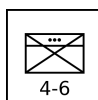
Infantry SAM Squad: Infantry SAM squads are equipped with shoulder-launched SAMs, with one launcher per squad. Their SAM capabilities and VP values are given elsewhere.

An infantry SAM squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.



Infantry Heavy SAM Squad: Infantry heavy SAM squads are equipped with pedestal-mounted SAMs, with one launcher per squad. Their SAM capabilities and VP values are given in a subsequent section.

An infantry heavy SAM squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.

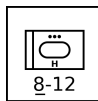


Infantry HQ Platoon: An infantry HQ platoon is the HQ unit of an infantry battalion or regiment. It consists of the formation commander, their staff, and their protection detail.⁴

ARMORED VEHICLE UNITS

Armored ground-combat vehicle units combine armored protection for their crew and weapons, mobility, and either a fire-power or transport capability.

Armored cars and armored half-tracks are considered to be light tanks, light anti-tank missile vehicles, light reconnaissance vehicles, light IFVs, or light or open APCs according to their role and armament. That is, the means of traction — tracked, half-tracked, or wheeled — is a secondary consideration.



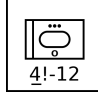
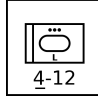
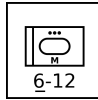
Tank Platoon: A tank platoon has armored vehicles principally armed with a direct-fire gun. The gun is most commonly mounted in a turret, but AFVs with casemate guns are also considered to be tanks. Its role is the support of infantry and the destruction of other tanks and armored vehicles. A tank platoon may have heavy, medium, light, or open armor.⁵

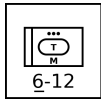
Heavy examples: M1 Abrams, Chieftain, Challenger, Leopard 2, and T-64/72/80.

Medium examples: M60, Centurion, Leopard 1, and T-54/55/62.

Light examples: M114A2, M551 Sheridan, Fox, Scimitar, Scorpion, and PT-76.

Open example: ASU-57.

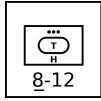
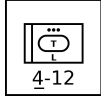




IFV Platoon: An infantry fighting vehicle (IFV) platoon is equipped with armored vehicles that transport infantry and provide them with fire support, being armed with a 20 mm or larger cannon and sometimes an anti-tank missile. Some IFVs are also designed to allow their infantry to fight while mounted. An IFV platoon may have medium or light armor.⁶

Medium example: M2A2 Bradley and Marder.

Light examples: M2/M2A1 Bradley, FV432 RARDEN, Warrior, and BMP-1/2.



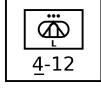
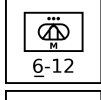
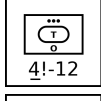
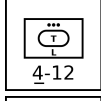
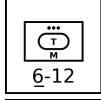
APC Platoon: An APC platoon is equipped with armored vehicles that transport infantry, but unlike IFVs are not equipped to provide fire support and at most are armed with a heavy machine gun. An APC platoon may have heavy, medium, light, or open armor. Heavy and medium APCs are typically converted tanks.⁷

Heavy examples: Nakpadon.

Medium examples: Nagmashot and Nagmachon.

Light examples: M113, FV432, Spartan, and BTR-50PK/60PB/70.

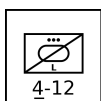
Open examples: M3 half-track and BTR-50P/60P.



Armored Anti-Tank Missile Platoon: An armored anti-tank missile platoon has armored vehicles equipped with anti-tank guided missiles. Its mission is the destruction of other armored vehicles. Conventional gun-armed tank destroyers such as the Kanonen Jpz are considered to be tanks. It is often a battalion- or regiment-level asset. An anti-tank platoon may have medium or light armor.⁸

Medium examples: Rakaten Jpz and Jaguar.

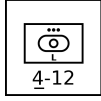
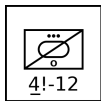
Light examples: M150, M901, FV438, Striker, and BRDM.



Armored Reconnaissance Platoon: An armored reconnaissance platoon has armored vehicles specialized in scouting and observation and with no offensive weapons more powerful than machine guns. (Vehicles used for reconnaissance that do have significant offensive weapons are typically classified as light tanks or IFVs.) A reconnaissance platoon may have light or open armor.⁹

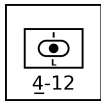
Light examples: M114, Ferret, and BRDM-2.

Open example: M20.



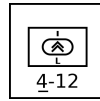
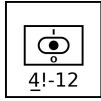
Armored Mortar Platoon: An armored mortar platoon has mortars mounted in armored vehicles (most commonly converted APCs or IFVs). It is typically a battalion-level asset in mechanized-infantry or tank battalions, and its role is indirect fire support. An armored mortar platoon has light armor.¹⁰

Examples: M106 and M125.



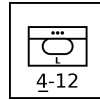
Armored Artillery Battery: An armored artillery battery has guns or howitzers in armored vehicles. It is typically a regiment-, brigade-, or division-level asset, and its role is indirect fire support. An armored artillery battery may have light or open armor.¹¹

Examples: M109, 2S3, and 2S1.



Armored Rocket Artillery Battery: An armored rocket artillery battery is the rocket-armed equivalent of an armored artillery battery. An armored rocket artillery battery has light armor.¹²

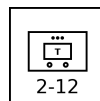
Example: M270.



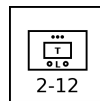
Armored HQ Platoon: An armored HQ platoon is the HQ unit of a tank, mechanized-infantry, or armored-artillery battalion, regiment, or brigade. It consists of the formation commander, their staff, and their protection detail. It is typically equipped with modified APCs or IFVs. An armored HQ platoon has light armor.

UNARMORED VEHICLE UNITS

Unarmored or soft vehicles provide mobility, but do not provide (or provide only incomplete) armor protection for their crew and weapons. They include armored vehicles whose crew or weapons are exposed. They may be wheeled or tracked.

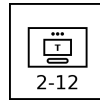


Truck Platoon: A truck platoon has unarmored wheeled vehicles that can transport medium or heavy loads of infantry, towed weapons, and supplies, including fuel and ammunition.



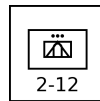
Light Truck Platoon: A light truck platoon has unarmored wheeled vehicles that can transport light loads of infantry, towed weapons, and supplies, including fuel and ammunition.

Examples: M151 Jeep, HMMWV, Landrover, Ilitis, and GAZ-69.



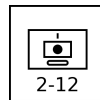
Tracked Carrier Platoon: A tracked carrier platoon is equipped with unarmored tracked vehicles that can transport medium loads of infantry, towed weapons, and supplies, including fuel and ammunition.¹³

Example: Bv 202.



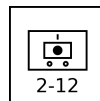
Mobile Anti-Tank Missile Platoon: A mobile anti-tank missile platoon is equipped with anti-tank missiles mounted on unarmored light trucks.

Examples: M151 TOW and M966 HMMWV TOW.



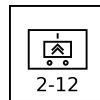
Tracked Artillery Battery: A tracked artillery battery has guns or howitzers mounted on tracked vehicles that provide mobility, but give only limited protection from small arms or artillery shrapnel. It is typically a regiment-, brigade-, or division-level asset, and its role is indirect fire support.¹⁴

Examples: M107, M110, 2S4, and 2S7.



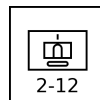
Mobile Artillery Battery: A mobile artillery battery has howitzers mounted on wheeled vehicles that provide mobility, but give only limited protection from small arms or artillery shrapnel. It is typically a regiment-, brigade-, or division-level asset, and its role is indirect fire support.¹⁵

Examples: CAESAR and Archer.



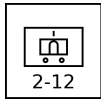
Mobile Rocket Artillery Battery: A mobile rocket artillery battery is the rocket-armed equivalent of a mobile artillery battery.¹⁶

Examples: M142 HIMARS and BM-21 Grad.



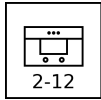
Tracked Missile Artillery Battery: A tracked rocket artillery battery is the rocket-armed equivalent of a mobile tracked battery.¹⁷

Example: M752 Lance.



Mobile Missile Artillery Battery: A mobile missile artillery battery is the missile-armed equivalent of a mobile artillery battery.¹⁸

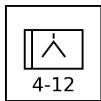
Examples: M1003 Pershing II and OTR-21 Tocha (SS-21).



Mobile HQ Platoon: A mobile HQ platoon is typically the HQ unit of a battalion, regiment, or brigade other than those of infantry, tank, mechanized infantry, or armored artillery. It is equipped with trucks, and its personnel are not specialized in fighting as infantry.

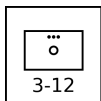
TOWED UNITS

Towed weapons provides neither inherent mobility nor protection. However, they are light and cheap compared to similar vehicle-mounted weapons. Towed weapons is typically used away from direct combat and provided with appropriate ground or helicopter transport.



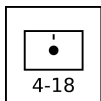
Towed Anti-Tank Gun Battery: A towed anti-tank gun battery has anti-tank guns on towed mounts with at most only very limited protection. It is often a regimental-, brigade-, or divisional-level asset.

Example: 2A18 (T-12) and 2A45 (Sprut).



Towed Mortar Platoon: A towed mortar platoon has 120 to 160 mm mortars on unprotected, towed mounts. It is often a regimental- or brigade- asset. Smaller infantry mortars are found in infantry mortar platoons.

Example: M1943.



Towed Artillery Battery: A towed artillery battery has guns or howitzers on towed mounts with at most only very limited protection. It is often a regimental-, brigade-, or divisional-level asset.

Examples: M119, M198, and 2A18 (D-30).

Ground-Combat and Transport Unit Table

Type	Size ^a	Defense Strength	Sighting Range	Mobility	VPs		AAA Class ^b
					3D/2D/D		
Infantry	Platoon	4	6	U	5/3/2		B2
Infantry Weapons	Platoon	3	6	U	6/4/2		B2
Infantry Mortar	Platoon	3	6	U	6/4/2		B2
Infantry FAC	Squad	5	6	U	-/-/8		—
Infantry SAM	Squad	5	6	U			—
Infantry Heavy SAM	Squad	5	6	U			—
Infantry HQ	Platoon	4	6	U	10/7/3		B2
Heavy Tank	Platoon	8	12	U	12/8/4		B3
Medium Tank	Platoon	6	12	U	10/7/3		B3
Light Tank	Platoon	4	12	U	6/4/2		B3
Open Tank	Platoon	4	12	U	5/3/2		B3
Medium IFV	Platoon	6	12	U	10/7/3		B3
Light IFV	Platoon	4	12	U	6/4/2		B3
Heavy APC	Platoon	8	12	U	10/7/3		B3
Medium APC	Platoon	6	12	U	8/5/3		B3
Light APC	Platoon	4	12	U	6/4/2		B3
Open APC	Platoon	4	12	U	5/3/2		B3
Medium Armored Anti-Tank Missile	Platoon	6	12	U	12/8/4		B3
Light Armored Anti-Tank Missile	Platoon	4	12	U	10/7/3		B3
Light Armored Reconnaissance	Platoon	4	12	U	6/4/2		B3
Open Armored Reconnaissance	Platoon	4	12	U	4/3/1		B3
Light Armored Mortar	Platoon	4	12	U	10/7/3		B3
Light Armored Artillery	Battery	4	12	U	12/8/4		B3
Open Armored Artillery	Battery	4	12	U	12/8/4		B3
Light Armored Rocket Artillery	Battery	4	12	U	12/8/4		B3
Light Armored HQ	Platoon	4	12	U	12/8/4		B3
Truck	Platoon	2	12	G/P	4/3/1		—
Light Truck	Platoon	2	12	G/P	4/3/1		—
Tracked Carrier	Platoon	2	12	U	4/3/1		—
Mobile Anti-Tank Missile	Platoon	2	12	G/P	8/5/3		—
Mobile Artillery	Battery	2	12	G/P	9/6/3		—
Tracked Artillery	Battery	2	12	U	10/7/3		—
Mobile Rocket Artillery	Battery	2	12	G/P	9/6/3		—
Tracked Missile Artillery	Battery	2	12	U	10/7/3		—
Mobile Missile Artillery	Battery	2	12	G/P	9/6/3		—
Mobile HQ	Platoon	2	12	G/P	10/7/3		—
Towed Mortar	Platoon	3	12	T	7/5/2		—
Towed Anti-Tank Gun	Battery	4	12	T	8/5/3		—
Towed Artillery	Battery	4	18	T	8/5/3		—

^a Platoons and batteries are available as sections and squads with corresponding modifications to their damage resiliences and VPs. ^b Platoons with AAA class of B2 and B3 can employ barrage fire to altitudes of 2 and 3 levels, respectively.

Vehicle Types Table

Heavy Tanks		Light Tanks		Heavy APCs		Light Armored Reconnaissance		Light Truck	
Al-Khalid/VT-1A	PK/CN 2001	AMX-10 RC	FR 1981	Achzarit	IL 1988	BRDM-2	SU 1973	GAZ-69	SU 1953
Ariete	IT 1995	AMX-13	FR 1953	Nakpadon	IL 1998	Fennek	DE/NL 2003	HMMWV	US 1985
Arjun	IN 2004	Fox	GB 1975	Namer	IL 2008	Ferret	GB 1952	Ilitis	GE 1978
Challenger 1	GB 1983	Commando (90 mm)	US 1964	Puma	IL 1993	M113 C&V/Lynx	US 1966	Landrover	UK 1949
Challenger 2	GB 1998	LAV I Cougar	CA 1976	Medium APCs		M114	US 1962	M38 1/4-ton truck	US 1949
Chieftain	GB 1967	LAV II Coyote	CA 1990	Boxer APC		Open Armored Reconnaissance		M151 1/4-ton truck	US 1960
Conqueror	GB 1955	M8/Greyhound	US 1943	Nagmachon		M20		UAZ-469	SU 1971
IS-2	SU 1944	M24 Chaffee	US 1944	Nagmashot				Willys MB 1/4-ton truck	US 1941
IS-3	SU 1945	M41 Walker Bulldog	US 1953	Light APCs		Light Armored Mortar		Tracked Carrier	
K1	KR 1987	M113 C&V with 25 mm	US 1974	Bronco/Warthog		FV432 with 81 mm		AT-T/S/L	
K2 Black Panther	KR 2014	M114A2	US 1969	BvS10/Viking		M106		Bv 202	
Leclerc	FR 1992	M551 Sheridan	US 1969	BvS10/VPK		M125		Bv 206	
Leopard 2	DE 1979	Panhard AML	FR 1961	BTR-50PK		M1064		BvS10/Beowulf	
M-84	YU 1985	Panhard EBR	FR 1951	BTR-60PA/PAI/PB		Stryker MCV		M548	
M1 Abrams	US 1980	PT-76	SU 1951	BTR-70		Light Armored Artillery		Mobile Anti-Tank Missile	
Merkava 1	IL 1979	Saladin	GB 1958	BTR-80		2S1 Gvozdika		M151 TOW	
Merkava 2	IL 1983	Scimitar	GB 1974	BTR-152K		2S3 Akatsiya		M966 HMMWV TOW	
Merkava 3	IL 1989	Scorpion	GB 1973	Commando		2S19 Msta-S		Mobile Artillery	
Merkava 4	IL 2004	SpPz Luchs	DE 1975	FV432		AMX-30 AuF1		Archer	
PT-91	PL 1995	Stryker MGS	US/CA 2006	LAV I Grizzly		AS-90		CAESAR	
Stridsvagn 103	SE 1967	Strv 74	SE 1958	LAV II Bison		Bkan 1		Tracked Artillery	
T-10	SU 1954	Type 62	CN 1961	LVTP-5		FV433 Abbot		2S4 Tyulpan	
T-14 Armata	RU 2024	Type 63	CN 1963	LVTP-7 (AAV)		L-33 Ro'Em		2S7 Pion	
T-64	SU 1966	Type 05 AAV	CN 2005	M59		M52		M41	
T-72	SU 1973	Type 08 ARV	CN 2008	M75		M108		M107	
T-80	SU 1976	Open Tanks		M113		M109		M110	
T-84	UA 1999	ASU-57	SU 1951	OT-64 SKOT		Panzerhaubitze 2000		AMX-13 F3	
T-90	RU 1992	SU-76	SU 1943	Piranha APC		Open Armored Artillery		Mobile Rocket Artillery	
Type 10	JP 2012	Medium IFVs		Saracen		AMX-105A		BM-8	
Type 90	JP 1990	Light IFVs		Saxon		M-50		BM-13	
Type 99	CN 2001	Medium Tanks		Spartan		M7 Priest		BM-31	
AMX-30		BMP-3		Stryker ICV		M40		BM-24	
ASU-85		Boxer IFV		TPz Fuchs		M44		BM-14	
Centurion		Strf 9040/CV90		Type 63		Light Armored Rocket Artillery		BM-21 Grad	
Comet		LAV III (add-on-armor)		Type 77		M270 MLRS		BM-27 Uragan	
Cromwell		M2A2/3/4 Bradley		Type 89		BM-1		BM-30 Smerch	
ISU-122		M3A2/3/4 Bradley		Type 08 APC				M142 HIMARS	
ISU-152		SPz Marder		VAB				Tracked Missile Artillery	
Kanonen JPz		SPz Puma		YP-408				M752 Lance	
Leopard 1		Light IFVs		Open APCs				AMX-30 Pluton	
M-50 Sherman		AIFV		AT-P				Mobile Missile Artillery	
M-51 Sherman		AMX-10P		BTR-40				9K52 Luna-M (Frog-7)	
M103		BMP-1		BTR-152A				Corporal	
M26 Pershing		BMP-2		BTR-50P				Sergeant	
M4 Sherman		Commando (20 mm)		BTR-60P				M34/M50 Honest John	
M46 Patton		FV432 RARDEN		M3 half-track				M74 Redstone	
M47 Patton		LAV III		SKP/VKPF				M790 Pershing 1a	
M48 Patton		LAV-25		SKPF/VKPF				M1003 Pershing II	
M60		M2 and M2A1 Bradley		Medium Armored Anti-Tank				OTR-21 Tochka (SS-21)	
Panzer 61		M3 and M3A1 Bradley		Raketen JPz 1				RSD-10 Pioneer (SS-20)	
Panzer 68		M113A1 FSC		Raketen JPz 2					
SU-100		Piranha IFV		Jaguar 1					
T-34		Pbv 301		Jaguar 2					
T-54		Pbv 302		Light Armored Anti-Tank					
T-55		SPz Kurz		BRDM-2 ATGM					
T-62		SPz Lang HS.30		BRDM-3 ATGM					
TAM		Stryker ICVD		Fennek MRAT					
Type 59		Type 86		FV438 Swingfire					
Type 61		Type 92		M150					
Type 69		Type 05 IFV		M901 ITV					
Type 74		Type 08 IFV		Striker					
Type 79		VCBI		Stryker ATGM					
Type 85		Warrior		VAB HOT					
Type 88		YPR-765		YP-408 PRAT					
Type 96				YPR-765 PRAT					
Type 15									
Vickers MBT									
Vijayanta									

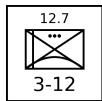
AAA UNITS

AAA units use guns to destroy attacking aircraft, but also have a secondary ground-combat capability. These units are presented below with their counters.

The AAA Unit Table summarizes the properties of AAA units. As well as the basic properties common to all ground units, the table gives: the AAA class (light, medium, or heavy); the maximum altitude; the limits of short, medium, and long range in hexes; the hit rolls at short, medium, and long range; the damage rating; whether the unit has an all-weather or ranging-only fire-control radar and its frequency band; whether it has night infrared sights; and whether it is also equipped with a SAM system. If the unit has a SAM, this will be described in detail in the section on SAM launcher units.¹⁹

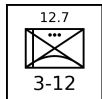
All AAA platoons and batteries are also available as sections (but not squads), with damage resiliences of 2, hit rolls reduced by 1, and VPs corresponding to a platoon or battery that has taken 1D damage.

LIGHT AAA UNITS

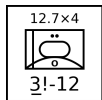


M2 Platoon: The Browning M2 .50 cal (12.7 mm) heavy machine gun is normally found on a low tripod for ground combat, but here is used on a tall dual-purpose mount suitable for air defense. A platoon has six guns.²⁰

The M2 gun was introduced in 1933 and has been used by United States forces and their allies in numerous conflicts since then.

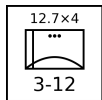


DShK-38 Platoon: The DShK-38 12.7 mm heavy machine gun is most commonly found on a low, wheeled mount for ground combat, but here is used on a tall tripod suitable for air defense. A platoon has six guns.²¹ The DShK-38 entered service with the Soviet Army in 1938 and saw action in World War II and numerous conflicts since then, including the Korean and Vietnam Wars, in which the Soviet Union participated or provided arms.

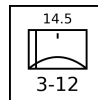


M16 Section: The M16 Multiple Gun Motor Carriage has an M45 turret with four M2 .50 cal (12.7 mm) heavy machine guns mounted on a modified M3 half-track. A section has two vehicles.²²

The M16 was used by United States forces during World War II and (primarily as an anti-personnel weapon) in the Korean War for airfield defense. Additionally, Israeli forces employed it during the 1956 Arab-Israeli War.

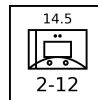
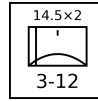


M55 Platoon: The M55 has an M45 turret with four M2 .50 cal (12.7 mm) heavy machine guns mounted on a towed carriage. A platoon has four gun mounts.²³ The M55 was used by United States forces during the Korean War and by Pakistani forces during the 1965 India-Pakistan War.



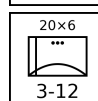
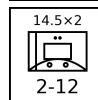
ZPU-1/2/4 Battery: The ZPU-1/2/4 have single, twin, or quad 14.5 mm heavy machine guns in open mounts on towed carriages. A battery has six gun mounts.²⁴²⁵²⁶

The ZPU-1/2/4 was introduced in 1949 and saw extensive service with Soviet or Soviet-supported forces in many subsequent conflicts, including with communist forces in the Korean War and Vietnam War and with Arab forces in the various Arab-Israeli Wars from 1956 onwards.



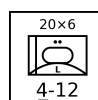
Mobile ZPU-1/2/4 Section: This is a section of two light trucks mounting ZPU-1/2/4 guns.²⁷²⁸

Many users of the ZPU-1/2/4 mount it on a suitable truck for increased mobility. Like the towed versions, it has seen combat in many conflicts.



M167 Platoon: The M167 Vulcan Air Defense System (VADS) has a 20 mm six-barreled Vulcan gun in an open mount on a towed carriage. The fire-control system has optical sights and a range-only radar. A platoon has four gun mounts.²⁹

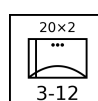
The M167 was introduced into service with the US Army in 1967, principally for airfield defense, and subsequently served with other US allies.



M163 Section: The M163 Vulcan Air Defense System (VADS) has a 20 mm six-barreled Vulcan gun in a lightly armored but open-topped turret on an adapted M113 APC. The fire-control system has optical sights and a range-only radar. A section has two vehicles.³⁰

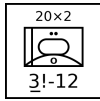
The M163 VADS was introduced into service with the US Army in 1968 and has since been adopted by other US allies. The M163 was used by US forces in the Vietnam War, the 1989 Invasion of Panama, and the 1991 Gulf War. It was also used by the Israel Army in the 1982 Lebanon War and in 2002 during the Second Intifada.

From 1988 in the US Army, each M163 was typically issued with a FIM-92 Stinger launcher with two rounds, so each M163 section effectively carried a mounted FIM-92 section.

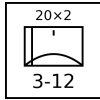


TCM-20 Platoon: The TCM-20 is an Israeli adaptation of the M55, with two Hispano-Suiza 20mm guns from obsolete aircraft replacing the four .50 cal machine guns. As with the M55, the fire-control system is entirely optical. A platoon has four gun mounts.³¹

The TCM-20 was employed by Israeli forces from 1970 and subsequently saw combat in the 1973 Arab-Israeli War and the 1982 Lebanon War.

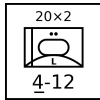


M3 TCM-20 Section: The M3 TCM-20 has a TCM-20 turret mounted in an adapted M3 half-track, like the earlier M16. A section has two vehicles.³² The M3 TCM-20 was used by Israeli forces in the 1973 Arab-Israeli War and the 1982 Lebanon War.

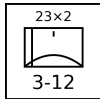


Twin Rh-202 Battery: The Rheinmetall Rh-202 has a single or twin 20mm gun in an open mount on a towed carriage. The fire-control system is entirely optical. The most common version is the twin mount. A battery has six gun mounts.³³

The twin version entered service with West German forces in 1972 and was used for airfield defense. Subsequently, it was exported and notably used by the Argentine Air Force in the 1982 South Atlantic War in the defense of BAM Malvinas (Port Stanley Airport) and BAM Condor (Goose Green Airfield).

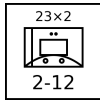


Panhard M3 DCA Section: The Panhard M3 DCA vehicles have an armored turret for twin 20 mm guns mounted on the Panhard M3 APC. The guns have optical sights, but typically one vehicle in each section or platoon is equipped with a search-and-ranging radar and can share this information automatically with the others. A section has two vehicles.³⁴ The Panhard M3 DCA served with the armed forces of Côte d'Ivoire, Niger, and the UAE.



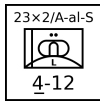
ZU-23 Battery: The ZU-23 has twin 23 mm cannons in an open mount on a towed carriage. It was designed as a more powerful replacement for the ZPU-1/2/4. The fire-control system is entirely optical. A battery has six gun mounts.³⁵

The ZU-23 was introduced into service by the Soviet Army in 1960. It later served with the armed forces of many Soviet allies, including with North Vietnam in the Vietnam War, with Egypt and Syria in the 1967 and 1973 Wars, with Soviet forces in the Soviet-Afghan War, with Iraq in the Iraq-Iran War, with Syria in the 1982 Lebanon War, and with both sides in the Afghan Civil War.



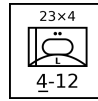
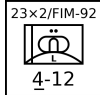
Mobile ZU-23 Section: This is a section of two light trucks mounting ZU-23 guns.³⁶

As with the earlier ZPU-1/2/4, many users of the ZU-23 mount it on vehicles, including trucks, tracked carriers, and APCs, for better mobility.



Sinai 23 Section: The Sinai 23 has a Dassault TA20 turret with two 23 mm guns and six Ayn-al-Saqr (similar to the Strela-2M) or Stinger missiles on an M113 chassis. It has night IR sights. A section has two vehicles. Each battery operates with one vehicle equipped with an RA20S search radar.³⁷

The Sinai 23 entered service with the Egyptian Army in 1991.



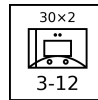
ZSU-23-4 Section: The ZSU-23-4 has four radar-controlled 23 mm guns, similar to those in the ZU-23, mounted in an armored turret on a hull adapted from the PT-76 light tank. A section has two vehicles.³⁸

The ZSU-23-4 entered service in the Soviet Army in 1965, replacing the ZSU-57-2 and outclassing all NATO anti-aircraft guns at that time. A platoon of four was assigned to the air-defense battery of tank and motor-rifle regiments and later complemented by a platoon of four SA-9 vehicles. It was exported to Soviet allies and saw extensive combat with Arab forces in the 1973 Arab-Israeli War and with Iraqi forces in the Iran-Iraq War and the 1991 Gulf War.



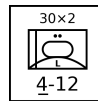
LAV-AD Section: The LAV-AD has a 25 mm GAU-12 rotary cannon and eight FIM-92D Stinger missiles mounted in a turret on the LAV-25 vehicle. It does not have radar, but is equipped with a passive night IR sight. A section has two vehicles.³⁹

The LAV-AD entered service with the USMC in 1997, but was retired after only a few years.



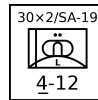
M53/59 Section: The M53/59 has a twin M53 30 mm gun mount on a partially armored truck. The fire-control system is entirely optical. A section has two vehicles.⁴⁰

The M53/59 entered service in the Czech Army in 1959 and was used instead of the ZSU-57-2. It was also exported to Iraq and saw combat in the Iran-Iraq War and the 1991 Gulf War.



AMX-30SA Section: The AMX-30SA is equipped with a pair of radar-controlled 30 mm Hispano-Suiza guns mounted in an armored turret on the hull of an AMX-30 tank. It is an improved version of the AMX-30 DCA. A section has two vehicles.⁴¹

The AMX-30SA served only in the Saudi Army from 1979.



Tunguska Section: The Tunguska has twin radar-controlled 30mm guns and four SA-19 missiles in an armored turret on an armored and tracked hull. It was designed to counter the A-10 and AH-64, against which the earlier ZSU-23-4 had weaknesses. A section has two vehicles.⁴²

The Tunguska entered service in the Soviet Army in 1984. It has been used by Russia, Ukraine, and Belarus after the break-up of the Soviet Union and has been exported.

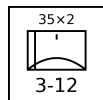
The Tunguska is referred to as the "ZSU-30-2" in *Air Strike*.



Tunguska-M Section: The Tunguska-M is an improved version of the Tunguska with eight missiles instead of four. A section has two vehicles.⁴³

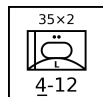
The Tunguska-M entered service in the Soviet Army in 1990. It has been used by Russia, Ukraine, and Belarus after the break-up of the Soviet Union and has been exported.

MEDIUM AAA UNITS



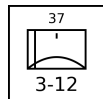
Oerlikon GDF Battery: The Oerlikon GDF has two 35 mm guns in an open mount on a towed carriage. A battery has six gun mounts. The basic fire-control system is optical, but a section is often coupled with the Super Fledermaus or Skyguard fire-control radars. (These radars cannot control a whole battery, only a section.)⁴⁴

The Oerlikon GDF entered service in 1963 and has served widely. Notably, it saw combat in the South Atlantic War with the Argentine forces in the defense of BAM Malvinas (Port Stanley Airport) and BAM Condor (Goose Green Airfield).



Gepard Section: The Gepard has two radar-controlled 35 mm guns in an armored turret on an adapted Leopard 1 tank hull. A section has two vehicles.⁴⁵

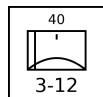
The Gepard entered service in 1976 with the West German Army and also served with the Belgian and Dutch armies. After the end of the Cold War, they were retired by these armies, and many were sold on to other countries.



61-K Battery: The 61-K (M1939) has a single 37 mm gun in an open mount on a towed carriage. The fire-control system is entirely optical, and (as an exception to the normal rule allowing medium AAA units to employ external FCRs) the 61-K cannot be coupled to an external fire-control radar. A battery has six gun mounts.⁴⁶

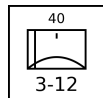
After entering service in 1939 with the Soviet Army, it served in WW2 and after until replaced by the S-60 57 mm gun. It also served with many Soviet allies, and in particular with communist forces in the Korean and Vietnam Wars.

The 61-K is referred to as the "M-38" in *Air Strike, Eagles of the Gulf*, and *The Speed of Heat*.⁴⁷



Bofors L/60 Battery: The Bofors L/60 has a single 40 mm gun in an open mount on a towed carriage. The fire-control system is entirely optical. A battery has six gun mounts.⁴⁸

It entered service in the 1940s and was widely used by Allied forces during WW2, both on land and sea. After the war, many continued to see service, although it was increasingly inadequate against faster jet aircraft and in many cases was replaced by the new L/70 model.



Bofors L/70 Battery: The Bofors L/70 has a single 40 mm gun in an open mount on a towed carriage. Between the 1930s and the end of WW2, aircraft speed increased substantially, and so the engagement time of the Bofors L/60 became inadequate. The new L/70 fired a lighter shell at a higher velocity and achieved a significant improvement in range and engagement time. The basic fire-control system is optical, but various external fire-control radars can be used with it, including the Super Fledermaus and Flycatcher systems. A battery has six gun mounts.⁴⁹

NATO forces and others adopted it widely starting in 1952.



Bofors L/70 BOFI and BOFI-R Battery: The BOFI version of the Bofors L/70 adds modern sights, a laser rangefinder, and proximity-fuzed shells. The BOFI-R version also adds an integrated fire-control radar. A battery has six gun mounts.⁵⁰

The BOFI is used by the armed forces of Brazil.⁵¹



S-60 Battery: The S-60 is a Soviet single 57 mm gun on an open mount on a towed carriage. The basic fire-control system is optical, but the gun is often used with the SON-9 (Fire Can) radar. A battery has six gun mounts.⁵²

It entered service in 1950 with Soviet forces as a replacement for the 61-K 37 mm gun. It was exported to many Soviet allies and saw combat with Arab forces in the 1967 and 1973 Arab-Israeli War, with communist forces in the Vietnam War, and with Iraqi forces in the Iran-Iraq War and the Gulf War. However, it was not used by communist forces in the Korean War. When paired with the SON-9 (Fire Can) radar, it was considered one of the more dangerous AAA systems faced by US aircraft in Vietnam.



ZSU-57-2 Section: The ZSU-57-2 has two 57 mm cannons in a lightly armored but open-topped turret on an armored and tracked hull. The cannons are similar to the S-60 cannons. The fire-control system is optical. A section has two vehicles.⁵³

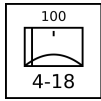
The ZSU-57-2 entered service in the Soviet Army in 1955 and replaced the BTR-40A and BTR-152A self-propelled 14.5 mm AA guns in the AA batteries of tank regiments. It was in turn replaced by the ZSU-23-4 from 1965. The ZSU-57-2 also served widely with Soviet allies and saw combat with Egypt and Syria in the 1967 and 1973 Wars, with Syria in the 1982 Lebanon War, with North Vietnam in the 1972 Easter Offensive and the 1975 Spring Offensive, and with various factions in the Yugoslav Wars in the 1990s.

HEAVY AAA UNITS



KS-12 Battery: The KS-12 (M1939 52-K) and the very similar KS-12A (M1944 KS-1) have a single 85 mm gun in an open mount on a towed carriage. The fire-control system is optical, but the guns are often paired with the SON-9 (Fire Can) radar. A battery has six gun mounts.⁵⁴

It entered service with Soviet forces in 1939. After WW2, it began to be replaced in Soviet service by the KS-19 100 mm and KS-30 130 mm guns, but was widely used by Soviet allies and saw combat with communist forces in the Korean War and the Vietnam War.



KS-19 Battery: The KS-19 has a single 100 mm gun in an open mount on a towed carriage. The fire-control system is optical, but like other contemporary Soviet guns, it was often paired with the SON-9 (Fire Can) radar. A battery has six gun mounts.⁵⁵

It entered service in 1948 with Soviet forces as a replacement for the KS-12 85 mm gun. It also served with many Soviet allies and saw combat with communist forces in the Korean War and Vietnam War and Iraqi forces in the Iran-Iraq and Gulf Wars.

AAA Units Table

Type	Size ^a	Year	Defense Strength		Sighting Range	Mobility	VPs		Gun	AAA Class	Altitude	Range			Hit			Damage Rating	FCR	Night IR Sights	SAM
							3D/2D/D					S	M	L	S	M	L				
M2	Platoon	1933	3	12	U		3/2/1		12.7 mm	L	5	2	3	4	3	2	1	1	—	—	—
DShK-38	Platoon	1938	3	12	U		3/2/1		12.7 mm	L	5	2	3	4	3	2	1	1	—	—	—
M16	Section	1944	3!	12	U		6/3		12.7 mm × 4	L	5	2	3	4	4	3	2	2	—	—	—
M55	Platoon	1945	3	12	T		4/3/1		12.7 mm × 4	L	5	2	3	4	5	4	3	2	—	—	—
ZPU-1	Battery	1949	3	12	T		3/2/1		14.5 mm	L	6	2	3	5	3	2	1	1	—	—	—
ZPU-2	Battery	1949	3	12	T		4/3/1		14.5 mm × 2	L	6	2	3	5	4	3	2	1	—	—	—
ZPU-4	Battery	1949	3	12	T		5/3/2		14.5 mm × 4	L	6	2	3	5	5	4	3	2	—	—	—
Mobile ZPU-1	Section	1949	2	12	G/P		2/1		14.5 mm	L	6	2	3	5	2	1	0	1	—	—	—
Mobile ZPU-2	Section	1949	2	12	G/P		3/2		14.5 mm × 2	L	6	2	3	5	3	2	1	1	—	—	—
Mobile ZPU-4	Section	1949	2	12	G/P		4/2		14.5 mm × 4	L	6	2	3	5	4	3	2	2	—	—	—
M167	Platoon	1967	3	12	T		7/5/2		20 mm × 6	L	8	2	4	6	6	5	4	3	R/HF	—	—
M163	Section	1968	4	12	U		8/4		20 mm × 6	L	8	2	4	6	5	4	3	3	R/HF	—	—
TCM-20	Platoon	1970	3	12	T		5/3/2		20 mm × 2	L	8	2	4	6	4	4	3	2	—	—	—
M3 TCM-20	Section	1970	3!	12	U		6/3		20 mm × 2	L	8	2	4	6	3	3	2	2	—	—	—
Single Rh-202	Battery	1972	3	12	T		4/3/1		20 mm	L	8	2	4	6	4	3	2	2	—	—	—
Twin Rh-202	Battery	1972	3	12	T		5/3/2		20 mm × 2	L	8	2	4	6	5	4	3	3	—	—	—
Panhard M3 DCA	Section	1975	4	12	U		7/4		20 mm × 2	L	8	2	4	6	4	4	3	2	R/UF	—	—
ZU-23	Battery	1960	3	12	T		6/4/2		23 mm × 2	L	9	2	5	8	5	4	3	3	—	—	—
Mobile ZU-23	Section	1960	2	12	G/P		5/3		23 mm × 2	L	9	2	5	8	4	3	2	3	—	—	—
Sinai 23 Ayn-al-Saqr	Section	1992	4	12	U		10/5		23 mm × 2	L	9	2	5	8	4	3	2	3	—	Y	Ayn-al-Saqr
Sinai 23 Stinger	Section	1992	4	12	U		12/6		23 mm × 2	L	9	2	5	8	4	3	2	3	—	Y	FIM-92
ZSU-23-4	Section	1965	4	12	U		9/5		23 mm × 4	L	9	2	5	8	5	4	3	3	W/VF	—	—
LAV-AD	Section	1997	4	12	U		14/7		25 mm × 5	L	10	3	6	9	4	3	2	4	—	Y	FIM-92D
M53/59	Section	1959	3	12	G/P		7/4		30 mm × 2	L	11	4	7	9	3	2	1	3	—	—	—
AMX-30SA	Section	1979	4	12	U		10/5		30 mm × 2	L	11	4	7	9	5	4	3	3	W/VF	—	—
Tunguska	Section	1984	4	12	U		20/10		30 mm × 2	L	11	4	7	9	5	4	3	4	W/VF	—	SA-19
Tunguska-M	Section	1990	4	12	U		22/11		30 mm × 2	L	11	4	7	9	5	4	3	4	W/VF	—	SA-19
Oerlikon GDF	Battery	1963	3	12	T		7/5/2		35 mm × 2	M	12	4	8	10	4	3	2	4	—	—	—
Oerlikon GDF	Section	1963	3	12	T		5/2		35 mm × 2	M	12	4	8	10	3	2	1	4	—	—	—
Gepard	Section	1976	4	12	U		11/6		35 mm × 2	M	12	4	8	10	5	4	3	4	W/VF	—	—
61-K	Battery	1939	3	12	T		5/3/2		37 mm	M	12	4	8	11	3	2	1	4	—	—	—
Bofors L/60	Battery	1935	3	12	T		5/3/2		40 mm	M	15	4	8	12	2	2	1	4	—	—	—
Bofors L/70	Battery	1952	3	12	T		6/4/2		40 mm	M	15	4	8	12	3	3	2	4	—	—	—
Bofors L/70 BOFI	Battery		3	12	T		8/5/3		40 mm	M	15	4	8	12	5	5	4	4	—	—	—
Bofors L/70 BOFI-R	Battery		3	12	T		9/6/3		40 mm	M	15	4	8	12	6	6	5	4	W/VF	—	—
S-60	Battery	1950	3	12	T		8/5/3		57 mm	M	18	5	10	15	2	2	1	5	—	—	—
ZSU-57-2	Section	1955	4!	12	U		6/3		57 mm × 2	M	18	5	10	15	2	1	1	5	—	—	—
KS-12	Battery	1939	4	18	T		9/6/3		85 mm	H	27	6	12	18	2	1	0	6	—	—	—
KS-19	Battery	1948	4	18	T		10/7/3		100 mm	H	39	7	14	21	1	1	0	6	—	—	—

^a Platoons and batteries are available as sections with corresponding modifications to their damage resiliences and hit rolls.

FCR UNITS

Most medium and heavy AAA batteries can be associated with an add-on fire-control radar (FCR), which improves their accuracy and allows them to engage unsighted targets at night and in adverse weather. These units are presented below with their counters.

The FCR Unit Table summarizes the properties of FCR units. As well as the basic properties common to all ground units, the table gives: the FCR class; the frequency band; and the modifier to the hit roll.



Towed FCR-A Platoon: FCR-As typically use 1940s technology. They include the SON-9 FCR.

SON-9: The Soviet SON-9 (Fire Can) FCR-A was derived from the earlier SON-4, which in turn was derived from the US-supplied SCR-584. It is commonly used with 57 mm S-60, 85 mm KS-12, 100 mm KS-19, and 130 mm KS-30 batteries. The SON-9 entered service in 1955 with Soviet forces and was widely exported to Soviet allies. It has seen combat, including with North Vietnam in the Vietnam War, Egypt and Syria in the 1967 War, the War of Attrition, and the 1973 War.⁵⁶



Towed FCR-B Platoon: FCR-Bs typically use 1950s technology.



Towed FCR-C Platoon: FCR-Cs typically use 1960s technology. They include the Super Fledermaus and Flycatcher FCRs.

Super Fledermaus: The Contraves Super Fledermaus FCR-C can control a section of two Oerlikon GDF 35 mm guns; it cannot control a whole battery. It entered service in 1965 with the Swiss Air Force and also was used by the Argentine Air Force, Brazil, Iran, Japan, and South Africa. It saw combat in the 1982 South Atlantic War in the defense of BAM Malvinas (Port Stanley Airport).⁵⁷

Flycatcher: The Signaal Flycatcher FCR-C can control up to six Bofors L/60 or L/70 guns or SAM launchers. Flycatcher entered service with the Royal Netherlands Air Force in 1979. It subsequently served with the Indian Armed Forces from 1985, the Irish Defense Forces from 2003, the Royal Netherlands Army from 1987, the Thai Armed Forces from 1981, Turkey, and Venezuela.⁵⁸



Towed FCR-D Platoon: FCR-Ds typically use 1970s technology. They include the Skyguard.

FCR Unit Table

Type	Size	Defense Strength	Sighting Range	Mobility	VPs	FCR		
						Class	Frequency	Modifier
Towed FCR-A	Platoon	2	12	T	6/4/2	A	LF	-1
Towed FCR-B	Platoon	2	12	T	6/4/2	B	MF	-2
Towed FCR-C	Platoon	2	12	T	7/5/2	C	VF	-2
Towed FCR-D	Platoon	2	12	T	8/5/3	D	MW	-3

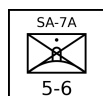
Skyguard: The Contraves Skyguard FCR-D can control two Oerlikon GDF 35 mm guns and additionally a SAM launcher with AIM-9 Sparrow, RIM-9 Sea Sparrow, or Aspide missiles. Skyguard entered service in 1977 with the Swiss Air Force. It was subsequently very widely exported and used by the Argentine Army, Austria, Bahrain, Bangladesh, Brazil, Canada, Chile, China, Cyprus, Egypt, Greece, Indonesia, Iran, Italy, South Korea, Kuwait, Malaysia, Oman, Pakistan, Saudi Arabia, Spain, Taiwan, and the UK. It saw combat with the Argentine Army in the 1982 South Atlantic War in the defense of BAM Malvinas (Port Stanley Airport) and BAM Condor (Goose Green Airfield).

INFANTRY SAM UNITS

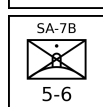
Infantry SAM units are squads of infantry specialized in the destruction of attacking aircraft with portable SAMs. These units are described below with their counters.

The Infantry SAM Unit Table summarizes the properties of infantry SAM units. In addition to the basic properties shared by all ground units, it gives: the number of ready, volley, and reload missiles; the multi-target capability; whether the unit has quick-reaction capability; whether the unit has IFF capability; whether the unit has night IR sights; and the optical lock-on number and range. Each squad has one launcher and, unless otherwise specified, one ready missile and two reload missiles. Infantry SAM squads may be attached to infantry, infantry weapons, infantry mortar, or infantry HQ platoons.

The Infantry SAM Table summarizes the properties of the actual missiles: guidance mode; launch roll; turn rate; flight time; visibility; ECCM, chaff, and flare ratings; whether the missile is active homing (AH) or has a home-on-jam (HOJ) capability; boost phase in FPs; base speed and sustainer duration in game turns; minimum altitude (and modifier for targets at T level); hit rolls for direct and proximity hits; and damage ratings for direct and proximity hits. Most infantry SAMs do not have true proximity fuzes, and so proximity hits in the game correspond to glancing direct hits in reality.

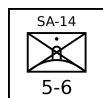


SA-7A/B Squad: The 9K32 Strela-2 (SA-7A Grail) was the first Soviet portable SAM. It has a rear-aspect, uncooled IR seeker.



The 9K32M Strela-2M (SA-7B Grail) is similar, but uses a rear-aspect, electrically-cooled IR seeker. In Warsaw Pact armies, the Strela-2M was sometimes used with the PRP portable RWR receiver that was capable of passively identifying enemy aircraft by their radar emissions.⁵⁹⁶⁰

The Strela-2 and -2M entered service with Soviet forces in 1968 and 1970. They have seen combat in almost every conflict since 1970 in which the Soviet Union or its successor states have been participants or supporters. In particular, the Strela-2M was used heavily by North Vietnamese forces in the 1973 Easter Offensive, by Arab forces in the 1973 Yom Kippur War, by Angolan forces in the South African Border War, by the Afghan Mujahedeen in the Soviet-Afghan War (having been supplied by the CIA), by Iran in the Iran-Iraq War in the 1980s, and by Serbian and Serbian-supported forces in the Yugoslav Wars in the 1990s.

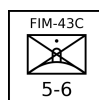


SA-14 Squad: The 9K34 Strela-3 (SA-14 Gremlin) is a development of the earlier Strela-2M. It has a rear-aspect, cryogen-cooled IR seeker. In Warsaw Pact armies, the Strela-3 was sometimes used with the PRP portable RWR receiver that was capable of passively identifying enemy aircraft by their radar emissions.⁶¹ The Strela-3 entered service with the Soviet Army in 1974. It has seen combat in many conflicts since then in which the Soviet Union or its successor states have been participants or supporters. In particular, it was used by Iraqi forces in the 1991 Gulf War.



SA-16/18 Squad: 9K310 Igla-1 (SA-16 Gimlet) and 9K38 Igla (SA-18 Grouse) use very similar missiles and launchers, but the Igla-1 has a single-channel IR seeker similar to the SA-14, and the Igla uses a dual-channel IR seeker to reduce its susceptibility to flares. The warhead was similar to the SA-7/14, but also exploded any remaining rocket fuel. In the game, this is simulated by increasing the damage rating by one if the missile attacks before expending more than half of its FPs (round down). Both incorporate IFF interrogators, but these were only used in Warsaw Pact armies.⁶²

The Igla-1 entered service with the Soviet Army in 1981 and the Igla in 1983. Both have been widely exported and have seen combat in many conflicts since then in which the Soviet Union or its successor states have been participants or supporters. The Igla was used by Iraqi forces in the 1991 Gulf War, by the Peruvian Army in the 1995 Cenepa War, and by Serbian forces in the 1999 Kosovo War.



FIM-43C Squad: The FIM-43C Redeye Block III was the first US infantry SAM to reach production and enter service. It has a rear-aspect, cryogen-cooled IR seeker.⁶³

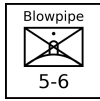
The FIM-43C entered service with the US Army in 1968 and subsequently was used by Australia, Chad, Denmark, West Germany, Greece, Israel, Jordan, Saudi Arabia, Somalia, Sudan, Sweden, and Turkey. It was also supplied by the CIA to the Afghan Mujahedeen and the Nicaraguan Contras.



FIM-92A/B/C/D Squad: The FIM-92A Stinger has an all-angle, second-generation, cryogen-cooled IR seeker. Its warhead weight 3 kg and was significantly more lethal than in earlier missiles. The FIM-92B Stinger POST adds a UV seeker to distinguish flares from aircraft. The FIM-92C/D Stinger RMP have software updates to further improve the performance against flares. All have an IFF interrogator to help avoid fratricide.⁶⁴

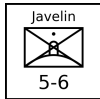


The FIM-92A entered service with the US Army in 1981, the FIM-92B in 1986, the FIM-92C in 1989, and the FIM-92D in 1992. They were exported to dozens of US allies including the Afghan Mujahedeen and the Angolan UNITA. They saw service in the 1982 South Atlantic War, the Soviet-Afghan War, the Angolan Civil War, the Libyan Invasion of Chad, the Tajik Civil War, the Chechen War, the Sri Lankan Civil War, the Syrian Civil War, and the Russian Invasion of Ukraine.



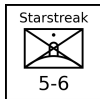
Blowpipe Squad: Blowpipe is an MCLOS optically-guided missile. While in theory this gave it an all-aspect capability, in practice guiding the missile to the target in combat proved to be very difficult.⁶⁵

Blowpipe entered service with the British Army and Royal Marines in 1975. It was also used by the Argentine Army, Canadian Army, Chilean Army and Navy, Ecuadorean Army, Guatemalan Army, Portuguese Army, Malawian Army, Malaysian Army, Nigerian Army, Omani Army, Royal Thai Air Force and Army, and the UAE Army. It saw combat in the 1982 South Atlantic War with both sides and had a low success rate. In 1986, it was trialed by the Afghan Mujahedeen, but again was not considered to be effective. Finally, it was also used in the 1995 Cenepa War by Ecuador.



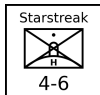
Javelin/Starburst Squad: Given the poor performance of Blowpipe in the 1982 South Atlantic War, Javelin was quickly developed as an upgraded version of Blowpipe with SACLOS guidance in place of MCLOS guidance. Starburst then replaced the radio-link with laser guidance. Javelin is also called "Javelin GL" and Starburst was originally called "Javelin S15".⁶⁶

Javelin replaced Blowpipe in the British Army and Royal Marines in 1984. It was also used by the Botswana Army, the Canadian Army, the Malaysian Army, the Peruvian Army, and the South Korean Army. Starburst in turn replaced Javelin in the British Army and Royal Marines in 1989. It was also used by the Canadian Army, the Malaysian Army, the Qatari Army, and the Royal Thai Army. Starburst served with the British Army in the 1991 Gulf War, but was not used in combat.



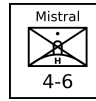
Starstreak Squad: Starstreak is a high-speed laser-guided missile. As the missile approaches the target, it launches three independent explosive darts. The darts have contact fuzes, but not proximity fuzes.⁶⁷

Starstreak replaced Javelin in the British Army and Royal Marines in 2000 and is also used by the South African Army and the Armed Forces of Ukraine. Starstreak served with the British Army in the 2003 Invasion of Iraq, but was not used in combat. It has seen combat in the Russian Invasion of Ukraine.



Starstreak LML Squad: The Starstreak Light Multiple Launcher (LML) has three ready-to-fire Starstreak missiles on a pedestal mount with a single sighting unit. The launcher and its associated equipment are heavy and will normally be transported by a vehicle unit.⁶⁸

Starstreak LML entered service with the British Army and Royal Marines in 2000. It is also used by Indonesia, Malaysia, and South Africa.



Mistral Squad: The Mistral is a French IR-homing missile on a pedestal mount. The missile has a considerably larger warhead than either the Stinger or Igla missiles, but this comes at a cost in portability. The launcher and its associated equipment are heavy and will normally be transported by a vehicle unit.⁶⁹

The Mistral entered service with the French Army in 1990. It was subsequently exported to many other countries.



RBS 70 Squad: The RBS 70 is a Swedish laser-guided missile on a pedestal mount. The launcher and its associated equipment are heavy and will normally be transported by a vehicle unit.⁷⁰

The RBS 70 entered service with the Swedish Army in 1977. It was subsequently used by Argentina, Australia, Bahrain, Brazil, the Czech Republic, Iran, Ireland, Lithuania, Norway, Pakistan, Singapore, Thailand, Tunisia, the UAE, Ukraine, and Venezuela.



Ayn-al-Saqr Squad: The Ayn-al-Saqr ("Eye of the Hawk") missile is a licensed version of the 9M32M Strela-2m (SA-7B Grail) manufactured in Egypt. It equips one version of the Sinai 23 air-defense vehicle.⁷¹

It entered service with the Egyptian Army in 1984.

Infantry SAM Unit Table

Name	Size	Year	Defense Strength	Sighting Range	Mobility	VPs		SAM	Missiles			Multi-Target	Quick Reaction	IFF	Night IR Sights	Lock-On		Other Names
						3D/2D/D			Ready	Volley	Reload					Optical	Range	
SA-7A	Squad	1968	5	6	U	-/-/3		SA-7A	1	1	2	-	Y	-	-	7	9	9K32 Strela-2 (Grail)
SA-7B	Squad	1972	5	6	U	-/-/4		SA-7B	1	1	2	-	Y	-	-	7	9	9K32M Strela-2M (Grail)
SA-14	Squad	1974	5	6	U	-/-/6		SA-14	1	1	2	-	Y	-	-	7	12	9K34 Strela-3 (Gremlin)
SA-16	Squad	1981	5	6	U	-/-/8		SA-16	1	1	2	-	Y	Y	-	7	12	9K310 Igla-1 (Gimlet)
SA-18	Squad	1983	5	6	U	-/-/10		SA-18	1	1	2	-	Y	Y	-	8	16	9K38 Igla (Grouse)
FIM-43C	Squad	1968	5	6	U	-/-/4		FIM-43C	1	1	2	-	Y	-	-	7	9	Redeye Block III
FIM-92A	Squad	1981	5	6	U	-/-/9		FIM-92A	1	1	2	-	Y	Y	-	8	12	Stinger
FIM-92B	Squad	1986	5	6	U	-/-/10		FIM-92B	1	1	2	-	Y	Y	-	8	12	Stinger POST
FIM-92C	Squad	1989	5	6	U	-/-/10		FIM-92C	1	1	2	-	Y	Y	-	8	12	Stinger RMP
FIM-92D	Squad	1992	5	6	U	-/-/10		FIM-92D	1	1	2	-	Y	Y	-	8	12	Stinger RMP
Blowpipe	Squad	1975	5	6	U	-/-/6		Blowpipe	1	1	2	-	Y	Y	-	7	9	
Javelin	Squad	1984	5	6	U	-/-/8		Javelin	1	1	2	-	Y	Y	-	7	9	Javelin GL
Starburst	Squad	1989	5	6	U	-/-/8		Starburst	1	1	2	-	Y	Y	-	7	9	Javelin S15
Starstreak	Squad	2000	5	6	U	-/-/12		Starstreak	1	1	2	-	Y	Y	-	7	12	
Starstreak LML	Squad	2000	4	6	U	-/-/14		Starstreak	3	1	0	-	Y	Y	-	7	12	
Mistral	Squad	1990	4	6	U	-/-/11		Mistral	1	1	2	-	Y	-	-	7	12	
RBS 70	Squad	1977	4	6	U	-/-/8		RBS 70	1	1	2	-	Y	Y	-	7	10	
Ayn-al-Saqr	Squad	1984	5	6	U	-/-/4		Ayn-al-Saqr	1	1	2	-	Y	-	-	7	12	

Infantry SAM Table

Name	Year	Guidance Mode	Launch Roll	Turn Rate	Flight Time	Visibility	ECCM	Chaff	Flare	AH	HOJ	Instant Arming	Boost Phase	Base Speed	Sustainer	Minimum Altitude	Hit		Damage		Other Names
																	Direct	Proximity	Direct	Proximity	
SA-7A	1968	I	6	BT	1	6	-	-	5	-	-	Y	-	10	0	T	5	6	4	2	9M32 Strela-2 (Grail)
SA-7B	1972	I	7	BT	1	6	-	-	5	-	-	Y	-	10	0	T	5	6	4	2	9M32M Strela-2M (Grail)
SA-14	1974	M	7	BT	1	6	-	-	4	-	-	Y	-	9	0	T	6	7	4	2	9M34 Strela-3 (Gremlin)
SA-16	1981	M	8	BT/2	1	6	-	-	4	-	-	Y	-	12	0	T	6	7	4*	2*	9K310 Igla-1 (Gimlet)
SA-18	1983	A	8	BT/2	1	6	-	-	2	-	-	Y	-	12	0	T	7	8	4*	2*	9K38 Igla (Grouse)
FIM-43C	1968	M	7	BT	1	6	-	-	5	-	-	Y	-	10	0	T	6	7	4	2	Redeye Block III
FIM-92A	1981	A	8	BT/2	1	2	-	-	3	-	-	Y	-	14	0	T	7	8	5	3	Stinger
FIM-92B	1986	A	8	BT/2	1	2	-	-	1	-	-	Y	-	14	0	T	7	8	5	3	Stinger POST
FIM-92C	1989	A	8	BT/2	1	2	-	-	1	-	-	Y	-	14	0	T	7	8	5	3	Stinger RMP
FIM-92D	1992	A	8	BT/2	1	2	-	-	1	-	-	Y	-	14	0	T	7	8	5	3	Stinger RMP
Blowpipe	1975	OG	8	HT	1	6	-	3	0	-	-	-	-	8	0	T	3	7	6	5	
Javelin	1984	OG	8	ET	1	5	-	2	0	-	-	Y	-	10	0	T	5	8	6	5	Javelin GL
Starburst	1989	LG	8	ET	1	5	-	1	0	-	-	Y	-	10	0	T	5	8	6	5	Javelin S15
Starstreak	2000	LG	8	ET	1	5	-	1	0	-	-	Y	-	24	0	T	7	9	6	4	
Mistral	1990	A	8	ET	1	5	-	-	2	-	-	Y	-	18	0	T	7	9	6	3	
RBS 70	1977	LG	8	ET	1	6	-	2	0	-	-	-	-	12	0	T	5	8	6	3	
Ayn-al-Saqr	1984	I	7	BT	1	6	-	-	5	-	-	Y	-	10	0	T	5	7	4	3	

EXAMPLE ORDERS OF BATTLE

This appendix contains a number of example orders of battle and their correspondence in game units.

The orders of battle for battalions are presented with the headquarters unit in the top row, any battalion-level assets on the next row or rows, and finally the line companies on the remaining rows. Within each row, infantry or towed units appear on the left and their transport units on the right.

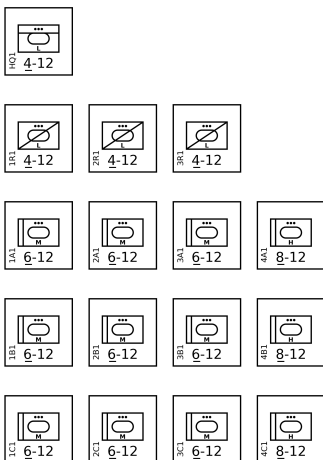
The unit identifiers for line platoons are the platoon number, followed by the company letter or number, followed by the battalion number. If only one battalion is used, the battalion number may be omitted. Transport platoons will have a "T" prefix. For example, "1B3" means the first platoon of B company of the third battalion, and "T1B3" is its transport platoon. Companies normally have letters in Western armies and numbers in Warsaw Pact armies. Additionally, AD is used for air defense, AR for artillery, AT for anti-tank, R for reconnaissance, and S for scout, support, or security. The unit identifiers for battalion HQ units are HQ followed by the battalion number.

Within the game, units are referred to as a squad, section, platoon/battery, or battalion according to their size. These terms are not used uniformly by all armies. For example, in armored formations in the British Army, the terms troop, squadron, and regiment are used in place of platoon, company, and battalion. Furthermore, even when these terms are used, they do not always refer to units of the same size. To help navigate these differences, when a different term is used for a formation, the game equivalent will appear in brackets.

BRITISH ARMY (EARLY 1960S)

The British Army in the early 1960s was in transition. It still operated with a mixture of medium Centurion, upgraded with the L7 105 mm gun, and heavy Conqueror tanks, and operated both mechanized and motorized infantry.⁷²

Armoured Regiment: This regiment [battalion] has three squadrons [companies] A/B/C each consisting of three tank troops [platoons] with Centurion medium tanks and one with Conqueror heavy tanks. The reconnaissance (R) squadron [company] is equipped with Ferret armored cars.

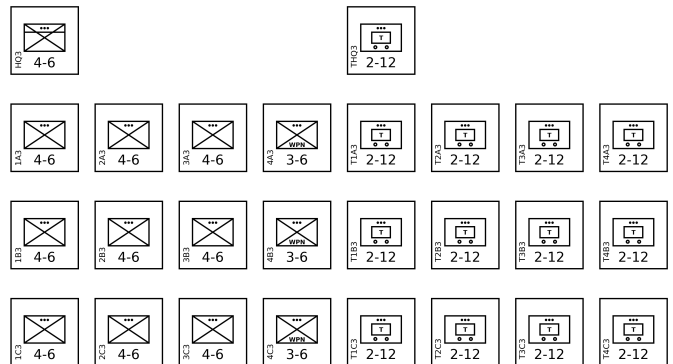


APC Infantry Battalion: This battalion has three companies A/B/C each consisting of three infantry platoons and one infantry

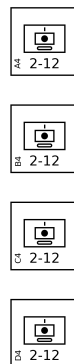
weapons platoon, with Saracen light APCs to provide mobility and protection.



Infantry Battalion: This battalion has three companies A/B/C each consisting of three infantry platoons and one infantry weapons platoon, with trucks to provide mobility.



Field Artillery Regiment: This regiment [battalion] has four batteries A/B/C/D each with six Cardinal (M44) self-propelled guns.



Heavy Artillery Regiment: This regiment [battalion] has four batteries A/B/C/D, of which A/B batteries [sections] each have two Honest John launchers and C/D batteries each have four towed 8-inch M1 guns with trucks.



BATTLE OF GOOSE GREEN

The 1982 Battle of Goose Green in the South Atlantic War was fought between 2 Para Battalion of the British Army and 12 Infantry Regiment of the Argentine Army, both with attached elements.⁷³

2 Para Battalion: 2 Para at the time consisted of three line companies A/B/D each with three platoons of infantry, one patrol company C with two infantry platoons, and a support company S with one platoon of mortars [but with only 2 out of 8 mortars available, so equivalent to a squad], one platoon of sustained-fire GPMGs, one platoon with MILAN portable ATGMs, and one engineers platoon. The attached units were half a battery [equivalent to a section] of three 105 mm Light Guns from Alma Battery of 29 Commando Regiment, Royal Artillery, two squads of Blowpipes from 43 Air Defence Battery, Royal Artillery, and a Recce Troop [platoon] of 59 Independent Commando Squadron, Royal Engineers.



Task Force Mercedes: The Argentine Army at Goose Green was called *Fuerza de Tareas Mercedes* (Task Force Mercedes). It consisted of 12th Infantry Regiment, with three line companies A/B/C, and reinforced with C company of 25th Infantry Regiment, three 105 mm pack howitzers [one section] of A Battery of 4th Airborne Artillery Regiment, an engineers platoon from the 9th Engineering Company, two Oerlikon GDF 35 mm AA guns with a Skyguard FCR of B Battery of 601st Anti-Aircraft Artillery Group, and a pair of SA-7B squads.



Military Airbase Condor: The Argentine Air Force renamed the airstrip at Goose Green as *Base Aérea Militar Cóndor* (Condor Military Air Base). The air-force elements at the base included a battery of two 20 mm Rh-202 AA guns of 3 Section B Battery 601AAAG with an Elta EWR and a security platoon.



ADDITIONAL RULES

RULE 1: GROUND UNITS

A ground *unit* is a combat or logistics formation of infantry, armor, transports, AAA, SAM launchers, or radars. This section discusses them in general.⁷⁴

A. Type and Size: A ground unit will have a type, for example, “heavy tank” or “ZSU-23-4”, and a size, which will be one of squad, section, platoon, or battery. Typically, a squad has five to ten soldiers and/or one vehicle, gun, or launcher; a section has ten to twenty soldiers and/or two vehicles, guns, or launchers; and a platoon or battery has thirty to fifty soldiers and/or three to six vehicles, guns, or launchers. The term “platoon” is more commonly used for infantry and tank units and “battery” for artillery and air-defense units, but within the game they are equivalent.

Companies, battalions, and other larger units above platoons or batteries are represented by multiple units. For example, an infantry company might be represented by three infantry platoon units.

All ground-combat and transport platoons are also available as sections and squads and all AAA platoons and batteries are also available as sections. A section or squad is identical in every way to a platoon or battery that has taken 1D or 2D damage, respectively, except that no VPs are awarded for this notional damage and non-AAA squads and sections may not engage in barrage fire.⁷⁵

In the game, ground units are uniformly referred to as squads, sections, platoons, and batteries. However, other terms are used for these units even within English-speaking armies. For example, in many Commonwealth armies a section is equivalent to a squad, a troop is equivalent to a platoon, and a squadron is equivalent to a company. In the US Armored Cavalry, a troop is equivalent to a company and a squadron is equivalent to a battalion.

B. Basic Properties: The type and size of a ground unit determines its basic properties: damage resilience, defense strength, target class, mobility, and sighting range.

The damage resilience is the amount of damage a ground unit can suffer before being killed. It is determined by the size of the unit: a squad has a damage resilience of 1D; a section has a damage resilience of 2D; and a platoon or battery has a damage resilience of 3D.⁷⁶

The numerical defense strength is used to calculate the odds for air-to-ground attacks and for ground-to-ground combat.

The target class is used to determine the appropriate attack strength of any weapons used in air-to-ground attacks. The class can be a hard, open, or soft according to the degree to which the unit enjoys armored protection. Many weapons have different attack strengths against targets of different classes. A hard target has armored protection from both direct (front, side, and rear) and top attacks. Hard targets are further described as light, medium, or heavy according to their degree of protection, and have corresponding defense strengths.⁷⁷ An open target has armored protection from direct attacks but not from top attacks. An open target is normally treated as a hard target, but against napalm, fire, FAE, and air-burst HE bombs, it is considered to be a soft target.⁷⁸ A soft target does not have armored protection their personnel and/or weapons.⁷⁹

C. Representation: The briefing documents show the graphical representation of ground units as counters.⁸⁰ The size of a unit is indicated by one, two, or three dots (for squads, sections, or platoons) or one vertical bar (for batteries) in an upper position. Hard targets are indicated by an underlined defense strength and open targets by an underline defense strength followed by an exclamation point.

Advanced Rule

D. Mobility: The mobility of a ground unit restricts the hexes it can occupy according to the terrain in and around those hexes. Infantry units, armored units (whether tracked or wheeled), and tracked unarmored units have unrestricted (U) mobility. Trucks and other wheeled unarmored units have good (G) or poor (P) mobility. All other units have towed (T) mobility.⁸¹

A unit with unrestricted mobility may occupy: any urban hex; any clear hex; or any forest hex. A unit with good mobility may occupy: any urban hex; any clear hex; or any forest hex with a road or trail. A unit with poor mobility may occupy: any urban hex; any clear or forest hex with a road, trail, or runway; or any clear hex adjacent to an urban hex or another clear hex with a road, trail, or runway. A unit with towed mobility may occupy hexes according to the mobility of its transport.⁸²

When necessary, a scenario must specify whether wheeled unarmored units have good or poor mobility, according to the terrain and the capabilities of the vehicles represented, and assign transport units to towed units.⁸³ A scenario may modify the mobility classes of ground units and the associated restrictions.⁸⁴

E. Composite Platoons: At the start of the game, an infantry, infantry weapons, infantry mortar, or infantry HQ platoon may attach one or two infantry squads to form a composite platoon. The attached squads are commonly infantry SAM or FAC squads. Once formed, a composite platoon may not be split into its constituent units.⁸⁵

An attached squad is not represented by a separate counter, does not count for stacking, and may not be sighted, identified, or attacked directly. Instead, it is considered to form part of the composite platoon. An attached squad retains its normal capabilities (for example, an attached SAM squad may engage aircraft and an attached FAC squad may sight and mark targets).

A composite platoon has the same damage resilience, defense strength, target class, mobility, and sighting range as its constituent platoon. If a composite platoon is sighted, no information is given on any attached squads. If a composite platoon is identified, any attached squads are also identified (for example, “infantry platoon with an attached FAC squad”). If a composite platoon is suppressed, each attached squad is suppressed too. If a composite platoon suffers damage, then for each D of damage, roll one die for each attached squad and on a 3– the attached squad is killed. If a composite platoon is killed, each attached squad is killed too. VPs are awarded for damage to each of the constituent units.⁸⁶

F. Transport: An infantry or towed unit may be transported by a specific associated transporting IFV, APC, truck, or tracked-carrier unit of at least the same size. When necessary, a scenario will establish the association between units and their specific transports.⁸⁷

A transported unit is not represented by a separate counter, does not count for stacking, and may not be sighted, identified, or attacked directly. It may not attack other ground units or aircraft, may not launch SAMs, may not use radar, and may not act as a FAC.

If a transporting unit is sighted, no information is given on the transported unit. If a transporting unit is identified, no information is given on a transported infantry unit, but any transported towed unit is also identified (for example, “truck platoon towing an artillery battery” or “truck platoon towing a ZPU-4 battery”). If a transporting unit suffers damage, then its transported unit suffers the same damage. VPs are awarded for damage to both the transporting and transported unit.

A unit may start the game either transported or not. If not transported, it may be placed according to either its mobility or that of its associated transport.⁸⁸

An infantry unit in the same hex as its associated transport may declare that it is mounting the transport at the start of a Ground Unit Interaction Phase. Both units must not be suppressed, and the transport must not have suffered more damage than the infantry unit. At the end of the Ground Unit Interaction Phase five game turns later, the unit is considered to be transported, and its counter is removed from the map.

An infantry unit being transported may declare that it is dismounting at the start of a Ground Unit Interaction Phase. The transport unit must not be suppressed. At the end of the Ground Unit Interaction Phase two game turns later, the unit is considered to be no longer transported, and its counter is placed on the map in the same hex as its transport. If placing the counter would violate stacking requirements, the unit must abort the dismounting process.

During these processes, the infantry unit may not perform any other action and the transport may not move and may not carry out ground attacks, but may perform barrage fire if it has this capability. The infantry unit may abort either of these process at the start of any Ground Unit Interaction Phase. If either are suppressed during the process, the unit must abort the process in the next Ground Unit Interaction Phase.⁸⁹

A towed unit may not mount or dismount during the game.

NOTES